

negative log-likelihood = loss

Gaussian noise
 $y \sim N(\hat{y}, \sigma^2)$

MLE

mean squared error
least squares

Bernoulli target
 $y \sim \text{Bern}(\hat{y})$

MLE

log loss
(binary cross entropy)

categorical target
 $\mathbf{y} \sim \text{Cat}(\hat{\mathbf{y}})$

MLE

categorical
cross entropy

negative log-prior = penalty

Gaussian prior
 $\theta \sim N(0, b^2)$

MAP

L2 penalty $\lambda \|\theta\|^2$
ridge

Laplace prior
 $\theta \sim \text{Laplace}(0, b)$

MAP

L1 penalty $\lambda \sum |\theta_j|$
lasso